Supplemental Table 1: Characteristics of Included Randomized Controlled Trials

Study,Year	Population	Inclusion Criteria	Type of Cardiac surgery	Definition of acute kidney injury	Events/Number of Patients (NAC)	Events/Number of Patients (Placebo)
Burns, 2005 [22]	Elective or urgent coronary artery bypass surgery patients with high risk for postoperative renal dysfunction	1 or more of the following criteria: - Pre-existing renal impairment (baseline serum creatinine level > 1.4 mg/dL [120 µmol/L]) - Ejection fraction - 35% grade 3 or 4 left ventricular function - Age 70 years - Diabetes mellitus - Concomitant valve or redo surgery	On pump	An increase in serum creatinine level greater than 0.5 mg/dL (44 µmol/L) or a 25% increase from baseline	44/148	42/145
Ristikankare, A. 2006 [18]	Patients with mild to moderate renal failure undergoing elective heart surgery with cardiopulmonary bypass	- preoperative plasma creatinine values above the upper range limit (>100 mmol litre^{-1})	On pump	An increase in serum creatinine level greater than 0.5 mg/dL (44 µmol/L) or a 25% increase from baseline	16/38	19/39
Haase, M 2007 [19]	Cardiac surgery patients at higher risk of postoperative renal failure	- Age of >70 years - Pre-existing renal impairment (preoperative serum creatinine level of > 120 mol/L) - New York Heart Association class III/IVb	On pump	A 25% increase in serum creatinine level from baseline	18/30	19/30

		- Valve surgery or complex cardiac surgery - Redo cardiac surgery - Insulin-dependent diabetes mellitus				
Adabag, A. S. 2008 [23]	patients with chronic kidney disease undergoing cardiac surgery	- Patients with pre-existing CKD, defined as estimated glomerular filtration rate (GFR) < 60 mL/(min)	On pump	An increase in serum creatinine level greater than 0.5 mg/dL (44 μmol/L) or a 25% increase from baseline	22/50	19/52
Sisillo, E. 2008 [24]	patients with chronic renal insufficiency (estimated creatinine clearance < or = 60 mL/min) undergoing elective cardiac surgery.	- chronic renal insufficiency (estimated creatinine clearance <60 mL/min)	On pump	A 25% increase in serum creatinine level from baseline	52/129	65/125
Prasad, A. 2010 [16]	Patients undergoing off-pump coronary artery bypass graft.	- age older than 70 years - diabetes mellitus - hypertension - baseline serum creatinine level higher than 133 μmol l^{-1} (1.5 mg dl^{-1}) - ejection fraction less than 35%.	Off pump	An increase in serum creatinine level greater than 0.5 mg/dL (44 μmol/L) or a 25% increase from baseline	3/35	4/35
Santana 2014 [17]	Patients with estimated glomerular filtration rate of 15 or greater and less than 60 mL · min−1 · 1.73 m−2	- Estimated glomerular filtration rate of 15 or greater and less than 60 mL min^{-1} 1.73 m^{-2}	On pump	AKIN	10/35	20/35

	scheduled to undergo elective CABG					
Song, J. W. 2015 [20]	Patients with known risk factors of AKI after off-pump coronary bypass surgery	1 of the following risk factors: - preoperative serum creatinine >1.4 mg/dl - left ventricular ejection fraction <35% or congestive heart failure - age >70 years - diabetes - re-operation.	Off pump	AKIN AKI was defined as an increase in serum creatinine more than or equal to 0.3 mg/dl from baseline, or to 50% from baseline, or an oliguria less than 0.5 ml/kg/h for more than 6 h within postoperative 48 h	19/60	20/57
Aldemir, M. 2016 [21]	Elderly patients undergoing coronary artery bypass grafting (CABG)	Geriatric	On pump	Plasma creatinine levels of >1.5mg/dl or a 25% increase from baseline	8/30	11/30
Amini 2018 [25]	Patients requiring elective off-pump coronary bypass graft surgery	AKIN congestive heart failure (CHF), active sepsis, preoperative ejection fraction lower than 40%, preoperative creatinine above 1.3 mg/dL	Off pump	AKIN AKI defined as a serum Cr increase of 0.3 mg/dL or ≥ 1.5 times baseline within 48 hours	15/68	10/71